A mechanics-constrained surrogate modelling framework for rapid structural response prediction with adaptive finite element data generation

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Abstract

Data-driven surrogates of finite element (FE) models can accelerate repeated structural assessment, but they usually return member forces that violate equilibrium and give no indication of when they should not be trusted. This paper presents a mechanics-constrained surrogate modelling framework for the linear response of planar moment frames, built and tested with the open-source PyNite FE library. A graph neural network ensemble predicts all nodal displacements and member forces in a dimensionless, load-case-separated form, so that superposition, homogeneity in the loads and in Young's modulus, and the boundary conditions hold exactly. An equilibrium projection with an uncertainty-weighted metric maps the predicted forces to the nearest statically admissible field, with a provable error reduction in that metric. Uncertainty estimates select the FE analyses used for training and refer doubtful predictions back to the solver. On 7,200 PyNite analyses, the surrogate reached median errors of 2.4% for end moments and 4.5% for horizontal displacements on in-distribution frames, four to six times lower than ridge regression, random forests, gradient boosting and a multilayer perceptron given the same features. On unseen geometries and load patterns the median moment errors were 7.5% and 8.7%, with heavy tails. The projection reduced nodal out-of-balance forces from 4–16% to machine precision. The ensemble spread ranked errors well for unseen geometries but not for unseen load patterns. Acquiring training frames by their equilibrium residual halved

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the FE budget needed on unseen geometries, whereas variance-based acquisition was worse than random sampling. The batched surrogate was 10–15 times faster than PyNite per frame, but for such cheap linear analyses its training cost is recovered only after many thousands of evaluations.

Keywords: Physics-informed machine learning, Surrogate model, Graph neural network, Equilibrium constraint, Deep ensemble, Active learning, Frame structures, Finite element analysis

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1. Introduction

Many tasks in structural engineering require the same structural system to be analysed many times: preliminary sizing and design-space exploration, code-based checks over large numbers of load combinations and variants, portfolio-scale assessment of existing buildings, optimisation and reliability analysis. Each analysis is cheap in isolation, but the number of repetitions makes finite element (FE) analysis the dominant cost of these loops. Surrogate models, which replace the FE solver by a fast approximation learned from a limited number of FE runs, have therefore a long history in engineering design [1] and reliability analysis [2, 3], and machine learning (ML) has substantially widened the range of quantities that can be emulated [4].

For structural assessment, however, a surrogate is only useful if its output can be trusted in the way an FE result is trusted. Three obstacles stand in the way. First, purely data-driven models carry no guarantee that their outputs are mechanically admissible: predicted member forces need not be in equilibrium with the applied loads, the predicted response need not scale linearly with the loads in a linear analysis, and boundary conditions are only approximately respected. An engineer who checks the reactions against the loads will immediately notice such defects. Second, accuracy reported on test data drawn from the training distribution says little about frames whose geometry or loading differ from those seen in training, which is the normal situation in practice. Third, the cost of generating training data is often treated as fixed, although the FE budget is usually the scarcest resource, and the surrogate itself does not indicate when its prediction should not be used.

Physics-informed machine learning [5] addresses the first obstacle by embedding physical knowledge in the learning model. Physics-informed neural networks penalise the residual of the governing equations in the loss [6] and

have been applied to solid mechanics [7]; in structural dynamics, physics-guided convolutional and recurrent networks have been used to model the seismic response of buildings [8, 9]. A penalty term, however, only encourages the constraint and leaves a residual violation whose size depends on the training. Constraints can instead be imposed exactly, either through the network output layer [10] or through hard-constraint formulations [11]. Graph neural networks (GNNs) [12, 13] are a natural representation for structures whose topology varies: they have been used to simulate particle and mesh-based physics [14, 15], to predict stress and deformation fields [16] and to build reusable surrogates for truss structures [17]. On the side of uncertainty, deep ensembles [18] provide a simple estimate of epistemic uncertainty, although its quality degrades under distribution shift [19]; and adaptive sampling driven by surrogate uncertainty is well established for Kriging models in structural reliability [2, 3] and more broadly as active learning [20].

What is missing is a workflow for skeletal structures that combines these elements so that (a) the predicted member forces are statically admissible by construction rather than on average, (b) the known linear structure of the problem is exploited instead of being learned, (c) the uncertainty estimate is used to decide both where to spend FE analyses during training and which predictions to hand back to the FE solver at deployment, and (d) the result is tested against conventional regression and ML baselines on geometries and load cases that the models have not seen. This paper develops and evaluates such a workflow for planar moment-resisting frames, using the open-source PyNite FE library [21] for every analysis. The research question is whether a carefully validated surrogate can predict displacements and member forces accurately enough to reduce repeated FE analysis.

The contributions are:

1. A mechanics-constrained graph surrogate in which load-case superposition, homogeneity in the load intensities and Young's modulus, and the boundary conditions hold exactly by construction, and in which exact statics quantities (storey shears, overturning moments, column gravity loads) and classical approximations (shear-building drift, beam-to-column stiffness ratio) are supplied as inputs.
2. An equilibrium projection layer that maps any predicted member-force field to the nearest statically admissible field, with an error-reduction guarantee in its own metric (Proposition 1) and a variance-weighted metric derived from the ensemble. The layer is model-agnostic and is

also applied to the baselines.

3. An uncertainty-guided workflow that uses the ensemble spread to select the FE analyses used for training and to refer unreliable predictions to the FE solver at deployment, with the cost–accuracy trade-off quantified by risk–coverage curves.
4. A benchmark of 7,200 PyNite frame analyses with in-distribution and two out-of-distribution test sets (unseen geometries; unseen load patterns), on which the workflow is compared with linear and quadratic ridge regression, random forests, gradient boosting and a multilayer perceptron. The code and data generator are released with the PyNite fork.

The remainder of the paper is organised as follows. Section 2 describes the frame family and the PyNite data generation, Section 3 the surrogate and the constraints, Section 4 the uncertainty-guided workflow and Section 5 the baselines and metrics. Results are presented in Section 6 and discussed, with the limitations of the study, in Section 7. Section 8 concludes.

2. Finite element data generation with PyNite

2.1. Parametric frame family

The study considers planar, multi-storey, multi-bay moment-resisting frames with rigid beam–column joints, a family that is routinely re-analysed during preliminary design and assessment. A frame is defined by the number of storeys n_s and bays n_b, the bay widths L_b, the storey heights h_s, Young's modulus E, the second moments of area of the columns ($I_{c,s}$) and beams ($I_{b,s}$) in each storey, the corresponding cross-sectional areas $A = I/r^2$ obtained from a sampled radius of gyration r, the base condition (fixed or pinned), the uniformly distributed gravity load $w_{s,b}$ on each beam and a lateral force F_s at each floor. All columns of a storey share one section and column stiffness may decrease with height through a taper factor. Units are kN and m throughout. The sampling ranges are listed in Table 1; E spans concrete-like to steel-like values so that the surrogate cannot rely on a fixed material.

Three scenarios are generated (Fig. 2):

- *Training design space (ID).* $1 \leq n_s \leq 5$, $1 \leq n_b \leq 4$; the gravity load is uniform along each floor but varies between floors; the lateral profile is either uniform or inverted-triangular, in either direction.

Table 1: Sampling ranges of the frame family. $\mathcal{U}$: uniform, $\log\mathcal{U}$: log-uniform. Units: kN, m.

Parameter	Training design space (ID)	Unseen scenarios
Storeys n_s	$\{1,\dots,5\}$	OOD-G: $\{6,7,8\}$ or $n_b \in \{5,6\}$
Bays n_b	$\{1,\dots,4\}$	OOD-G: up to 6
Bay width L_b	$\mathcal{U}(3,9)$	same
Storey height h_s	$\mathcal{U}(2.8,4.5)$	OOD-G: first storey $\mathcal{U}(4.5,6.0)$ in 50%
Young's modulus E (kN/m^2)	$\log\mathcal{U}(2.5\times10^7, 2.1\times10^8)$	same
Column I_c at base (m^4)	$\log\mathcal{U}(5\times10^{-5}, 2\times10^{-3})$, taper $\mathcal{U}(0.5,1)$	same
Beam $I_{b,s}$ (m^4)	$\log\mathcal{U}(5\times10^{-5}, 2\times10^{-3})$ per storey	same
Radius of gyration r	$\mathcal{U}(0.08,0.30)$ (columns, beams)	same
Base condition	fixed (70%) / pinned (30%)	same
Gravity load w (kN/m)	$\mathcal{U}(5,40)$ per floor, all bays	OOD-L: 30–60% of w in random bays
Lateral profile	uniform or inverted triangle	OOD-L: parabolic or roof-only
Roof lateral force (kN)	$\mathcal{U}(5,80)$, random sign	same

- *Unseen geometries (OOD-G).* Frames with 6–8 storeys and/or 5–6 bays, and in half of the cases a tall first storey (4.5–6.0 m), all outside the training ranges.
- *Unseen loading (OOD-L).* Training geometries subjected to pattern gravity loading, in which the live part of the load is randomly removed from individual bays, and to lateral profiles that were never seen in training (parabolic, or a single force at the roof).

The OOD-G set tests extrapolation in topology and size, the OOD-L set tests extrapolation in the load pattern; neither is used for training, model selection or early stopping.

2.2. PyNite analyses and statics verification

Each frame is built as an `FEModel3D` in PyNite [21] (version 3.2.0 of the fork cited) with one Euler–Bernoulli member per column and beam segment. Out-of-plane translations and rotations are restrained at every node so that the 3D solver reproduces planar behaviour. The lateral and gravity loads are defined as two separate load cases, W and G, and solved by a single first-order linear analysis (`analyze_linear`). For each case the nodal displacements (u_x, u_y, θ_z) and the global member end forces are extracted; the

latter are converted to the basic forces of Section 3.1. The wall-clock time of the complete PyNite call chain (model construction, analysis and result extraction) is recorded for every frame and later used for the cost comparison.

As a guard against convention errors, every FE result is checked against an independent implementation of the frame statics (Section 3.1): the out-of-balance nodal force computed from the PyNite member forces must vanish. All 7,200 analyses pass with residuals at round-off level (below 10^{-6} of the largest applied load; typically 10^{-11} kN). The data comprise a candidate pool of 5,000 ID frames, 400 ID validation frames, and three test sets of 600 frames each (ID, OOD-G, OOD-L), generated with disjoint random seeds.

3. Mechanics-constrained surrogate

The surrogate maps the frame description to the full response field: all nodal displacements and the basic forces of all members. Three kinds of mechanics are built into it, in decreasing order of strictness: (i) exact constraints that hold for any network output (boundary conditions, homogeneity in the load intensities, nodal equilibrium after projection); (ii) mechanics-derived inputs that are exact statical quantities or classical approximations; and (iii) an inductive bias in the architecture (message passing on the structural graph).

3.1. Statics of the frame and the equilibrium constraint

For member e between nodes i and j with length L_e and direction cosines (c_e, s_e), the basic forces are $\mathbf{q}_e = [N_e, M_{i,e}, M_{j,e}]^\mathsf{T}$: the axial force (tension positive) and the two end moments acting on the member. For a member carrying a transverse distributed load w_e, the six local end forces are

$$\mathbf{f}_e = \mathbf{b}_e \mathbf{q}_e + \mathbf{f}_e^0, \qquad \mathbf{b}_e = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1/L_e & 1/L_e \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -1/L_e & -1/L_e \\ 0 & 0 & 1 \end{bmatrix}, \tag{1}$$

where $\mathbf{f}_e^0 = -\frac{1}{2} w_e L_e \, [0, 1, 0, 0, 1, 0]^\mathsf{T}$ is the simply supported particular solution. This parametrisation satisfies member equilibrium identically [22].

Rotating to global axes and assembling at the nodes gives the nodal equilibrium equations at the n_f free degrees of freedom (DOFs),

$$\mathbf{C}\,\mathbf{q} = \mathbf{d}, \qquad \mathbf{C} \in \mathbb{R}^{n_f \times 3m}, \quad \mathbf{d} = \mathbf{P}_f - \mathbf{p}_f^0, \tag{2}$$

where $\mathbf{q}$ stacks the basic forces of all m members, $\mathbf{P}_f$ are the applied nodal loads and $\mathbf{p}_f^0$ the assembled particular solutions. $\mathbf{C}$ is the equilibrium (statics) matrix. For a kinematically stable frame it has full row rank and $3m-n_f$ equals the degree of static indeterminacy, so Eq. (2) defines an affine set of dimension $3m - n_f > 0$ that contains the exact FE solution but also every other statically admissible force field. Equilibrium alone therefore cannot determine the forces, which is precisely why a learned prior is needed; conversely, any force prediction that violates Eq. (2) is mechanically inadmissible. Support reactions follow from $\mathbf{q}$ at the restrained DOFs, so a force field satisfying Eq. (2) also satisfies global equilibrium. Assembling $\mathbf{C}$ costs $\mathcal{O}(m)$ and requires no stiffness matrix.

3.2. Dimensional analysis and exact load-case superposition

For a linear elastic frame the response is linear in the loads and inversely proportional to E. The surrogate enforces this exactly through dimensional analysis [23]. With the mean storey height H and the reference rigidity $EI_{\mathrm{ref}} = E\,\overline{I_c}$, each load case $c \in \{\mathrm{W}, \mathrm{G}\}$ is scaled by its own intensity, the signed base shear $\lambda_{\mathrm{W}} = \sum_s F_s$ and the total gravity load $\lambda_{\mathrm{G}} = \sum_e |w_e| L_e$, and the response is written as

$$\begin{aligned} (u_x, u_y) &= \sum_c \frac{\lambda_c H^3}{EI_{\mathrm{ref}}}\,(\hat{u}_{x,c}, \hat{u}_{y,c})(\boldsymbol{\pi}), \quad & \theta_z &= \sum_c \frac{\lambda_c H^2}{EI_{\mathrm{ref}}}\,\hat{\theta}_c(\boldsymbol{\pi}), \\ N &= \sum_c \lambda_c \hat{N}_c(\boldsymbol{\pi}), & M &= \sum_c \lambda_c H\,\hat{M}_c(\boldsymbol{\pi}), \end{aligned} \tag{3}$$

where hats denote the network outputs and $\boldsymbol{\pi}$ is a set of dimensionless input groups: lengths divided by H, sections divided by I_{ref} (areas as AH^2/I_{ref}), lateral loads divided by λ_{W} and gravity loads by λ_{G}. Because $\boldsymbol{\pi}$ is invariant to the load magnitudes and to E, Eq. (3) is exactly homogeneous of degree one in each load-case intensity and of degree -1 in E, and reversing the lateral load reverses the lateral response exactly. Predicting the two cases separately also avoids a severe scaling problem: in the combined case the lateral sway is often orders of magnitude smaller than the gravity-induced quantities, so

a single scale under-weights it. The ablation in Section 6.4 quantifies this choice. Restrained DOFs are multiplied by zero, so the boundary conditions also hold exactly.

3.3. Mechanics-informed features

Each node carries 25 dimensionless features and each member 10. Beyond position, topology and section ratios, they include quantities from elementary statics that are exact and cost $\mathcal{O}(n)$: the storey shear $V_s = \sum_{k \geq s} F_k$, the overturning moment of the loads above the storey, and the gravity load carried by each column line (tributary, exact for simply supported beams). Two classical approximations are added: the beam-to-column stiffness ratio of each storey, $\rho_s = (n_b I_{b,s}/\bar{L}_b)/((n_b+1) I_{c,s}/h_s)$, and the shear-building drift estimate $\delta_s = V_s/k_s$ with $k_s = (n_b + 1)\, 12EI_{c,s}/h_s^3$ (or $3EI/h^3$ for a pinned first storey), together with its height-wise cumulative sum. The network learns the correction from these simple estimates to the FE response, which is a much smoother target than the response itself. The same features are given to all baselines.

3.4. Graph network

Frames of different size and topology are represented as graphs whose nodes are joints and whose edges are members, and processed with an encode–process–decode message passing network [12, 13, 14, 15]. Node and edge features are encoded by two-layer perceptrons (width 96, SiLU, layer normalisation). Each of the eight processing steps updates every edge from its own state and the states of its end nodes, $\mathbf{z}_e \leftarrow \mathbf{z}_e + \phi_e([\mathbf{z}_e, \mathbf{h}_i, \mathbf{h}_j])$, and every node from the sums of the messages of the members that start and end at it, kept separate so that the network can distinguish the two ends of a member, $\mathbf{h}_i \leftarrow \mathbf{h}_i + \phi_n([\mathbf{h}_i, \sum_{e:i=\text{start}} \mathbf{z}_e, \sum_{e:i=\text{end}} \mathbf{z}_e])$. Eight steps nearly cover the graph diameter of the training frames (at most nine members), and frame-level information is supplied in addition through the statics features. Decoders output six dimensionless displacements per node and six dimensionless basic forces per member (three per load case). The loss is the mean squared error of the outputs standardised per component, over free DOFs only. Training uses AdamW [24] with a one-cycle learning-rate schedule (peak 2×10^{-3}), batches of 32 frames, gradient clipping at 1.0 and 150 epochs, implemented in PyTorch [25]. A variant with a soft equilibrium penalty $\|\mathbf{C}\hat{\mathbf{q}}_c - \mathbf{d}_c\|^2$ added to the loss is examined in the ablation.

3.5. Equilibrium projection

A data-driven prediction $\hat{\mathbf{q}}$ will in general violate Eq. (2). The proposed constraint layer replaces it with the closest statically admissible force field in a metric $\boldsymbol{\Sigma}^{-1}$:

$$\mathbf{q}^\star = \arg\min_{\mathbf{Cq}=\mathbf{d}} (\mathbf{q}-\hat{\mathbf{q}})^\mathsf{T}\boldsymbol{\Sigma}^{-1}(\mathbf{q}-\hat{\mathbf{q}}) = \hat{\mathbf{q}} + \boldsymbol{\Sigma}\mathbf{C}^\mathsf{T}\left(\mathbf{C}\boldsymbol{\Sigma}\mathbf{C}^\mathsf{T}\right)^{-1}(\mathbf{d}-\mathbf{C}\hat{\mathbf{q}}), \quad (4)$$

which follows from the Karush–Kuhn–Tucker conditions of the equality-constrained quadratic programme [26]. It is applied separately to each load case in dimensionless form. Two metrics are compared: the Euclidean metric on the dimensionless forces ($\boldsymbol{\Sigma} = \mathbf{I}$), and the *variance-weighted* metric $\boldsymbol{\Sigma} = \mathrm{diag}(\boldsymbol{\sigma}^2)$ built from the ensemble variance of each predicted force component (Section 3.6), so that the components the model is least sure of absorb most of the correction. If $\hat{\mathbf{q}}$ is read as the mean of a Gaussian belief with covariance $\boldsymbol{\Sigma}$, Eq. (4) is the exact posterior mean after conditioning on the equilibrium equations.

Proposition 1. *Let* $\mathbf{q}_{\mathrm{FE}}$ *be the exact solution, so that* $\mathbf{Cq}_{\mathrm{FE}} = \mathbf{d}$*. For any prediction* $\hat{\mathbf{q}}$ *and any symmetric positive definite* $\boldsymbol{\Sigma}$*, the projected forces satisfy* $\|\mathbf{q}^\star - \mathbf{q}_{\mathrm{FE}}\|^2_{\boldsymbol{\Sigma}^{-1}} = \|\hat{\mathbf{q}} - \mathbf{q}_{\mathrm{FE}}\|^2_{\boldsymbol{\Sigma}^{-1}} - \|\hat{\mathbf{q}} - \mathbf{q}^\star\|^2_{\boldsymbol{\Sigma}^{-1}} \le \|\hat{\mathbf{q}} - \mathbf{q}_{\mathrm{FE}}\|^2_{\boldsymbol{\Sigma}^{-1}}$.

Proof. $\mathbf{q}^\star$ is the $\boldsymbol{\Sigma}^{-1}$-orthogonal projection of $\hat{\mathbf{q}}$ onto the affine set $\mathcal{A} = \{\mathbf{q} : \mathbf{Cq} = \mathbf{d}\}$. Since $\mathbf{q}^\star - \mathbf{q}_{\mathrm{FE}}$ lies in $\ker\mathbf{C}$ and $\hat{\mathbf{q}} - \mathbf{q}^\star = \boldsymbol{\Sigma}\mathbf{C}^\mathsf{T}\boldsymbol{\mu}$ for some $\boldsymbol{\mu}$, their $\boldsymbol{\Sigma}^{-1}$-inner product is $\boldsymbol{\mu}^\mathsf{T}\mathbf{C}(\mathbf{q}^\star - \mathbf{q}_{\mathrm{FE}}) = 0$, and the identity follows from Pythagoras' theorem. □

The projection can therefore never increase the error measured in its own metric, and it removes exactly the part of the error that is visible to statics. It does not guarantee a reduction of every individual component (for example the end moments alone), which is why the choice of metric matters and is examined empirically. The projection needs one dense solve of size n_f per load case, which for the frames considered here takes a fraction of a millisecond, and it is model-agnostic: it is applied below to the baselines as well.

3.6. Predictive uncertainty

Epistemic uncertainty [27] is estimated with a deep ensemble [18] of $M = 5$ networks that differ in initialisation and batch order and share the training data and normalisation. The ensemble mean is the prediction and

the ensemble variance σ^2 of each output feeds the projection metric. For decisions at the level of a whole frame, the relative spread of the end moments,

$$s = \frac{\|\boldsymbol{\sigma}_M\|_2}{\|\hat{\mathbf{M}}\|_2}, \tag{5}$$

is used, where $\boldsymbol{\sigma}_M$ collects the physical standard deviations of all end moments. A second, label-free indicator is also examined: the equilibrium residual $\|\mathbf{C}\hat{\mathbf{q}} - \mathbf{d}\|/\|\mathbf{d}\|$ of the unprojected ensemble mean, which measures how far the raw prediction is from being statically admissible.

4. Uncertainty-guided workflow

The surrogate is embedded in a workflow with two uses of the uncertainty estimate (Fig. 1).

Adaptive FE data generation. Training starts from N_0 randomly chosen frames of the candidate pool, analysed with PyNite. In each round an ensemble is trained, every remaining candidate is scored without any FE analysis, the B highest-scoring candidates are analysed with PyNite and added to the training set. This is pool-based active learning [20] in the spirit of adaptive surrogate methods in structural reliability [2, 3]. Three acquisition rules are compared: random sampling, the ensemble variance (mean normalised variance over all outputs), and the equilibrium residual of the raw prediction. The candidates are labelled in advance for convenience, but a label is revealed to the learner only when the strategy requests that analysis, and the recorded PyNite time of each requested analysis is charged to the budget.

FE referral at deployment. For a batch of new frames, all are first predicted and projected. Frames whose score s (Eq. 5) is among the largest fraction α are sent to PyNite and their surrogate results discarded; the remaining predictions are accepted. The fraction α (or an equivalent threshold on s) sets the trade-off between cost and the worst accepted error, which is evaluated with risk–coverage curves [28].

5. Baselines and evaluation protocol

Baselines. Five baselines represent conventional regression and widely used machine learning: ridge regression [29] on the features (linear) and on all their

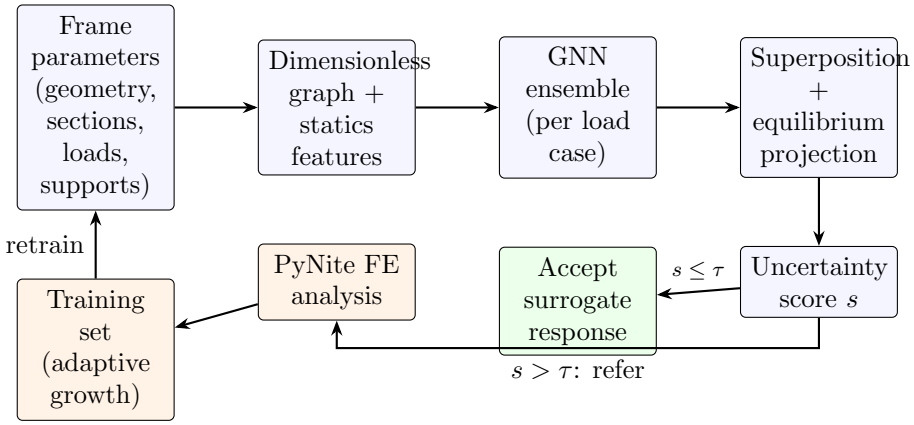


Figure 1: Proposed workflow. Exact mechanics (superposition, boundary conditions, equilibrium) is imposed around a learned graph ensemble; the ensemble spread decides which frames are analysed with PyNite, both while the training set is built and at deployment.

pairwise products (quadratic), random forests [30] (200 trees), histogram gradient boosting [31] (600 iterations, one model per output) and a three-layer perceptron (256 units per layer, early stopping), all from scikit-learn [32]. Because fixed-length regressors cannot ingest graphs of varying size, they predict per node (displacements, from the 25 node features) and per member (basic forces, from the 10 member features concatenated with the features of both end nodes). They receive the same dimensionless, superposition-split targets and the same hard boundary conditions as the graph network, so that the comparison isolates the learning model and the constraint layer rather than the preprocessing. The quadratic ridge model is fitted on a random subset of 30,000 rows to keep the design matrix in memory. All models are trained on the same first 3,000 frames of the ID pool.

Metrics. All errors are computed per frame in physical units against PyNite and then summarised over the test set by the median and the 95th percentile. They are the relative L_2 errors of the horizontal displacements of all nodes (e_{u_x}), of all member end moments (e_M) and of all axial forces (e_N); the relative error of the largest inter-storey drift ratio (e_{drift}) and of the largest absolute end moment ($e_{M,\max}$), which are the quantities that usually govern serviceability and member design; and the relative nodal out-of-balance force $\|\mathbf{r}\|/\|\mathbf{d}\|$ with $\mathbf{r} = \sum_c \lambda_c(\mathbf{C}\hat{\mathbf{q}}_c - \mathbf{d}_c)$.

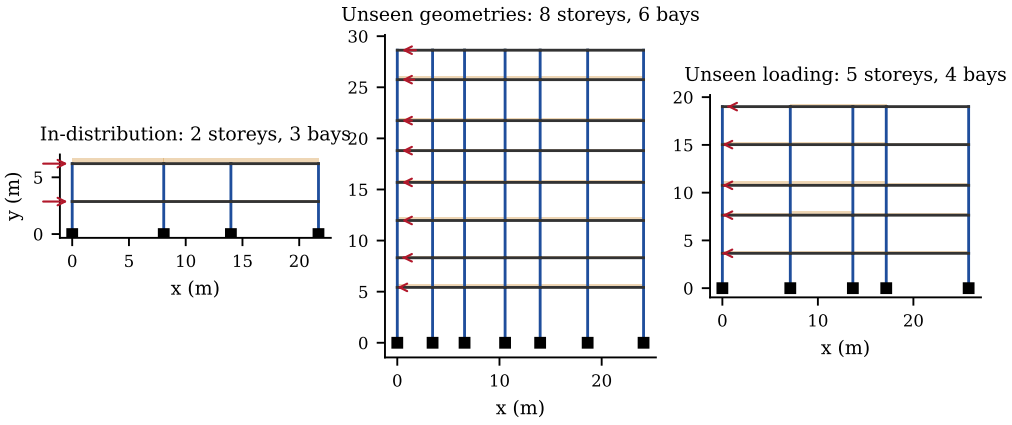


Figure 2: Example frames from the three test sets. Blue: columns; black: beams; shaded: gravity line loads (scaled); red arrows: lateral floor forces; squares and triangles: fixed and pinned bases. The unseen-geometry frame is taller and wider than any training frame; the unseen-loading frame carries pattern gravity loads and a parabolic lateral profile.

6. Results

All numbers in this section are produced by the released scripts from a single dataset and a single training run of each model, except the adaptive-sampling study, which is repeated with two random seeds. Errors are medians over 600 test frames unless stated otherwise.

6.1. Dataset

The 5,000-frame pool contains frames with 3 to 45 members (18 on average), the OOD-G set frames with 11 to 104 members (56 on average). A PyNite analysis of one frame (both load cases, including model construction and result extraction) took 37.5 ms on average for the ID frames and 104.7 ms for the larger OOD-G frames on the CPU used here (Section 6.7). The median largest inter-storey drift ratio is 0.46% in the ID test set and 0.79% in the OOD-G set; 15% and 25% of the frames, respectively, exceed 2%, because sections and loads are sampled independently and are not sized for serviceability. Since the surrogate and the baselines all predict dimensionless responses that are exactly invariant to the load intensities and to E (Section 3.2), the relative errors reported below would be identical for any rescaling of the loads, so the presence of very flexible frames does not bias the comparison.

6.2. Accuracy on seen and unseen frames

Table 2 compares the proposed surrogate (the five-network ensemble with variance-weighted equilibrium projection) with the five baselines, and Fig. 3 shows the full error distributions. On the in-distribution test set the proposed surrogate reaches median errors of 4.5% for the horizontal displacement field, 4.9% for the governing drift ratio, 2.4% for the end-moment field, 0.6% for the axial forces and 0.8% for the largest end moment of the frame. The best baseline, gradient boosting, has 29.0%, 30.2%, 10.0%, 3.2% and 3.4% respectively, so the graph surrogate is about four times more accurate for the member forces and six times more accurate for the displacements, although the baselines receive the same mechanics-informed features, targets and boundary conditions. The gap is largest for displacements, which depend on the stiffness of the whole frame and not only on local quantities; this is the information that message passing over the structural graph supplies and that per-node regression cannot reconstruct from local features. Linear and quadratic ridge regression perform worst, which shows that the input–output map is strongly nonlinear even though the structural response is linear in the loads.

All models lose accuracy on the unseen frames, but the ranking is unchanged. For unseen geometries (frames up to eight storeys and six bays, against at most five and four in training) the proposed surrogate has a median moment error of 7.5% and a median displacement error of 11.4%, against 22.5% and 43.1% for gradient boosting; the governing drift error is 13.7%, whereas the tabular baselines have errors of 72–143%. For unseen load patterns the median moment error is 8.7% (gradient boosting 17.3%, MLP 16.8%) and the displacement error 5.6%. The global quantities that usually govern design are predicted more accurately than the full fields: the largest end moment of a frame is within 2.2% (OOD-G) and 2.5% (OOD-L) of PyNite at the median. The tails, however, are heavy: the 95th percentile of the moment error is 22.6% for unseen geometries and 21.4% for unseen loading, which is not acceptable for design without further checks and motivates the referral mechanism of Section 6.5. Figure 4 shows the corresponding parity plots for the roof displacement and the largest end moment, and Fig. 5 the end moments of a typical unseen-geometry frame.

6.3. Equilibrium and the projection layer

Without the projection, every model produces member forces that are not in equilibrium with the applied loads (Fig. 6). The median out-of-balance

Table 2: Prediction errors on the three test sets (medians over 600 frames, in %, except the 95th percentile of e_M and the equilibrium residual). e_{u_x}: horizontal displacements; e_{drift}: largest inter-storey drift ratio; e_M: end moments; e_N: axial forces; $e_{M,\max}$: largest end moment; last column: median relative nodal out-of-balance force. Bold: best value per column and test set.

Model	e_{u_x}	e_{drift}	e_M	e_M (p95)	e_N	$e_{M,\max}$	Out-of-balance
In-distribution (600 frames)							
Linear ridge	53.5	56.0	38.1	61.5	20.6	12.3	4.3×10^{-1}
Quadratic ridge	51.0	66.0	18.4	40.4	5.7	4.7	1.0×10^{-1}
Random forest	33.6	40.5	16.9	31.1	6.9	5.0	1.4×10^{-1}
Gradient boosting	29.0	30.2	10.0	22.3	3.2	3.4	7.4×10^{-2}
MLP	39.5	59.9	10.8	25.4	4.9	3.4	1.2×10^{-1}
GNN, single (raw)	6.8	7.5	3.7	7.7	2.6	1.3	6.7×10^{-2}
GNN ensemble (raw)	**4.5**	**4.9**	2.5	**5.9**	1.8	0.8	4.2×10^{-2}
GNN ensemble + Eucl. proj.	**4.5**	**4.9**	4.0	9.5	1.1	1.2	$< 10^{-12}$
Proposed: GNN ens. + var.-weighted proj.	**4.5**	**4.9**	**2.4**	6.5	**0.6**	**0.8**	$< 10^{-12}$
Unseen geometries (600 frames)							
Linear ridge	61.4	72.3	82.2	192.6	36.2	35.9	1.2×10^{0}
Quadratic ridge	66.6	141.2	44.6	116.0	11.3	9.9	4.0×10^{-1}
Random forest	41.9	92.6	32.5	55.9	15.9	8.2	3.4×10^{-1}
Gradient boosting	43.1	118.3	22.5	41.1	6.2	5.8	1.8×10^{-1}
MLP	48.5	143.3	22.1	40.4	10.0	5.8	3.4×10^{-1}
GNN, single (raw)	14.6	26.9	11.0	23.2	7.5	3.5	2.3×10^{-1}
GNN ensemble (raw)	**11.4**	**13.7**	8.3	22.7	5.3	2.3	1.6×10^{-1}
GNN ensemble + Eucl. proj.	**11.4**	**13.7**	12.6	31.1	2.5	3.9	$< 10^{-12}$
Proposed: GNN ens. + var.-weighted proj.	**11.4**	**13.7**	**7.5**	**22.6**	**1.5**	**2.2**	$< 10^{-12}$
Unseen loading (600 frames)							
Linear ridge	52.8	57.5	44.3	83.6	26.2	15.2	5.2×10^{-1}
Quadratic ridge	53.2	75.8	26.7	49.0	9.4	7.5	2.1×10^{-1}
Random forest	33.8	42.9	24.2	38.2	12.0	7.0	2.3×10^{-1}
Gradient boosting	28.4	32.3	17.3	30.8	4.8	4.4	1.2×10^{-1}
MLP	38.0	50.1	16.8	31.4	6.4	4.5	1.5×10^{-1}
GNN, single (raw)	8.1	9.2	9.9	22.7	4.4	2.9	9.4×10^{-2}
GNN ensemble (raw)	**5.6**	**5.7**	9.1	21.5	3.5	**2.4**	6.7×10^{-2}
GNN ensemble + Eucl. proj.	**5.6**	**5.7**	10.8	23.7	2.2	3.1	$< 10^{-12}$
Proposed: GNN ens. + var.-weighted proj.	**5.6**	**5.7**	**8.7**	**21.4**	**1.6**	2.5	$< 10^{-12}$

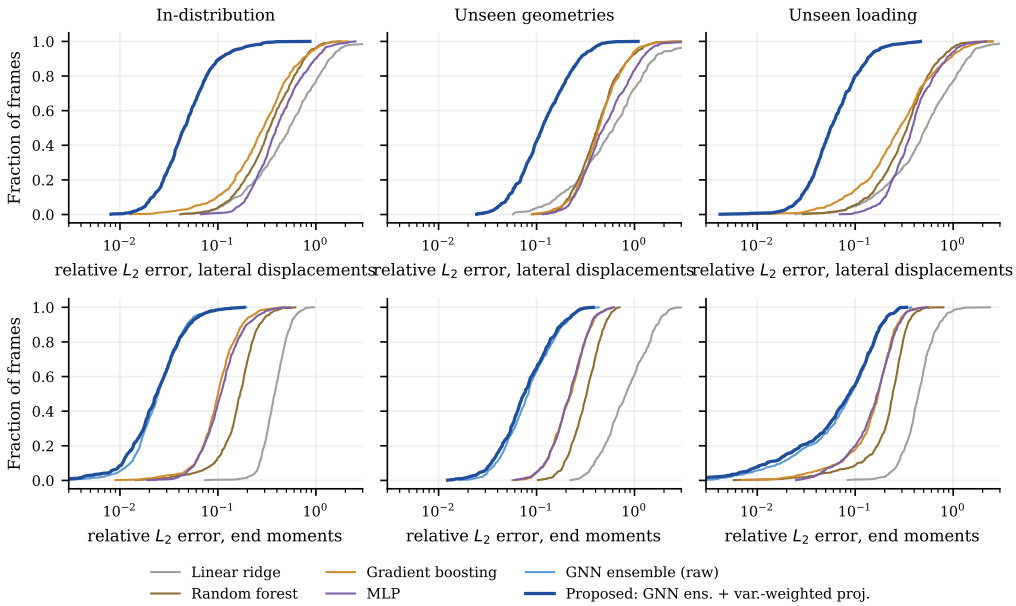


Figure 3: Cumulative distributions of the per-frame relative errors of the horizontal displacement field (top) and of the end-moment field (bottom).

force is 4% of the load vector for the raw graph ensemble and 7–43% for the baselines on in-distribution frames, rising to 16% and 18–120% for unseen geometries. A soft equilibrium penalty in the training loss barely changes this (Table 4): the residual of a single network falls from 6.7% to 6.3%. After projection (Eq. 4) the residual is at machine precision ($\approx 10^{-16}$) for every frame and every model, as guaranteed by construction, so the support reactions balance the loads and every nodal free body is in equilibrium.

The projection also changes the accuracy, and the metric matters. With the variance-weighted metric the median moment error of the ensemble decreases from 2.5% to 2.4% (ID), from 8.3% to 7.5% (OOD-G) and from 9.1% to 8.7% (OOD-L), and the axial-force error decreases by a factor of 2–3.5 (1.8%→0.6%, 5.3%→1.5%, 3.5%→1.6%), because axial forces are largely fixed by statics once the moments are known. The 95th percentile of the moment error is essentially unchanged on the unseen sets and increases slightly on the ID set (5.9%→6.5%). With the Euclidean metric, by contrast, the projection degrades the moments (2.5%→4.0% on ID, 8.3%→12.6% on OOD-G) while still improving the axial forces. This is consistent with Proposition 1:

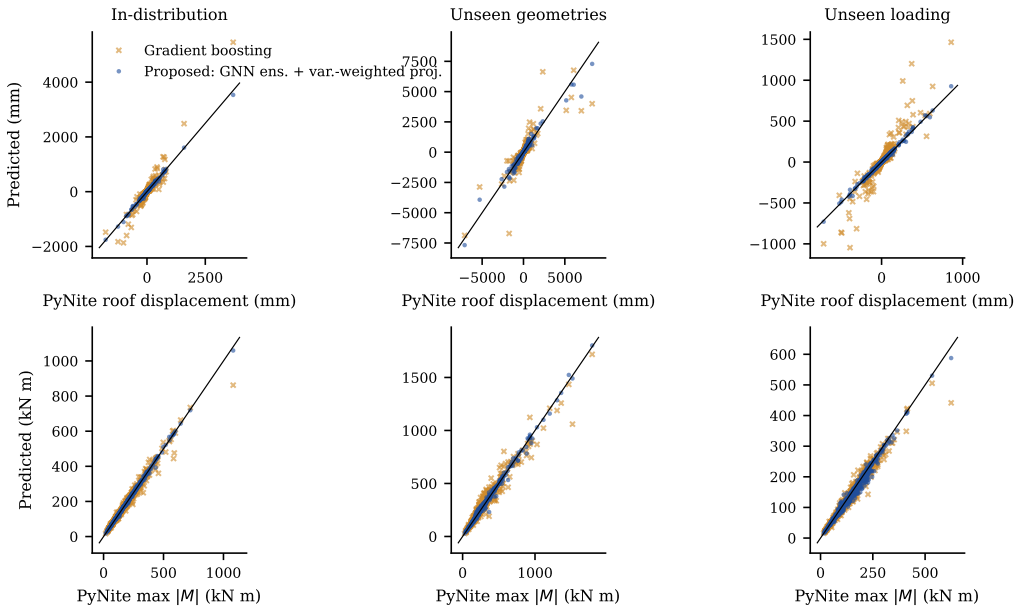


Figure 4: Predicted versus PyNite roof displacement (top; mean of the roof nodes) and largest absolute end moment (bottom) for the proposed surrogate and the best baseline.

the error can only decrease in the metric of the projection, and the Euclidean metric on dimensionless forces lets the correction flow into the moments, whose uncertainty it ignores. Weighting by the ensemble variance directs the correction to the least certain components.

Because the layer is model-agnostic, it was also applied to the baselines (Table 3). The baselines provide no per-output variance, so the Euclidean metric is used. The axial-force errors decrease for every baseline and test set, typically by a factor of about two. The moment errors decrease for gradient boosting (10.0%→9.1% ID, 22.5%→21.4% OOD-G, 17.3%→16.5% OOD-L) and for the MLP on two of the three sets, but increase for the linear ridge model and the random forest. Thus the projection always delivers admissible forces and always improves the axial forces, while its effect on the moments depends on whether the metric reflects where the prediction errors actually are.

6.4. Ablation of the mechanics constraints

Table 4 isolates the constraints with single networks. Replacing the exact load-case superposition by a single combined load scale roughly doubles the

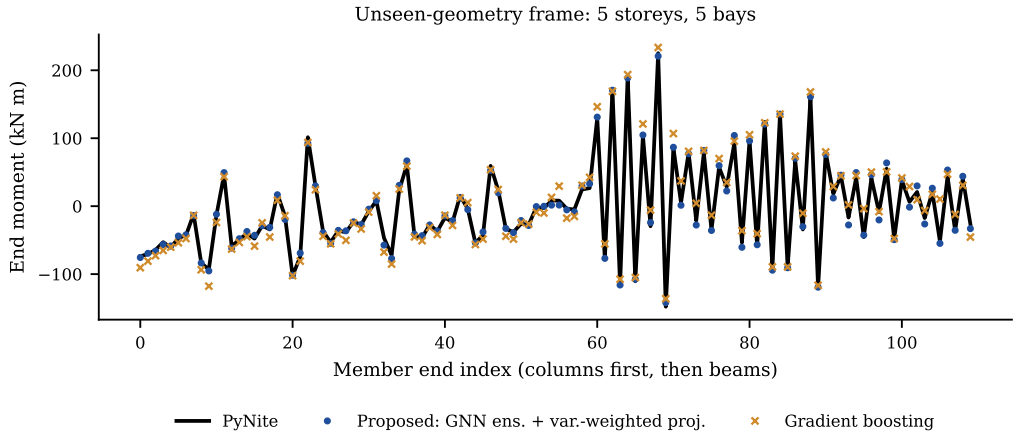


Figure 5: End moments of all members of an unseen-geometry frame (the OOD-G frame with the median number of members): PyNite, proposed surrogate and gradient boosting.

Table 3: Effect of the Euclidean equilibrium projection on the baselines: median moment error e_M and axial-force error e_N (%) before → after projection.

	In-distribution				Unseen geometries				Unseen loading			
Model	e_M	→	e_N	→	e_M	→	e_N	→	e_M	→	e_N	→
Linear ridge	38.1	51.8	20.6	11.3	82.2	108.5	36.2	17.7	44.3	55.9	26.2	13.1
Random forest	16.9	18.2	6.9	4.2	32.5	33.7	15.9	6.2	24.2	26.4	12.0	5.8
Gradient boosting	10.0	9.1	3.2	2.2	22.5	21.4	6.2	4.2	17.3	16.5	4.8	3.4
MLP	10.8	10.5	4.9	2.5	22.1	27.0	10.0	5.0	16.8	16.5	6.4	3.5

displacement errors (6.8%→12.2% ID, 14.6%→24.5% OOD-G, 8.1%→15.1% OOD-L) and increases the moment errors on all sets (3.7%→5.4% ID, 11.0%→15.0% OOD-G); the governing-drift error on unseen geometries rises from 27% to 69% (all metrics are in the released `results/main.json`). This confirms the scaling argument of Section 3.2: in the combined case the lateral sway is masked by the much larger gravity scale. The soft equilibrium penalty neither improves accuracy nor materially reduces the residual at the weight used, which supports enforcing equilibrium exactly instead of through the loss. Finally, ensembling five networks reduces the median moment error from 3.7% to 2.5% and the displacement error from 6.8% to 4.5% on the ID set (Table 2), in addition to providing the uncertainty estimate.

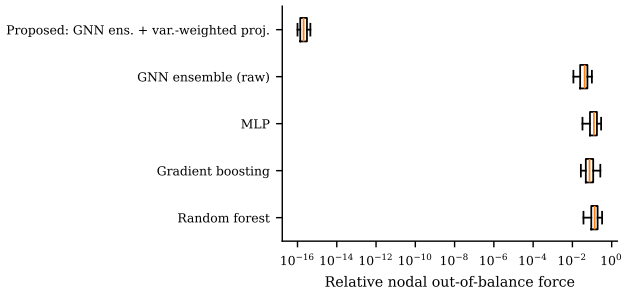


Figure 6: Relative nodal out-of-balance force of the predicted member forces on the in-distribution test set (box: quartiles; whiskers: 5th–95th percentiles). Only the projected surrogate is statically admissible.

Table 4: Ablation with single networks (same seed, data and training budget). Median e_{u_x} and e_M (%) and median relative out-of-balance force.

	In-distribution			Unseen geometries			Unseen loading		
Variant	e_{u_x}	e_M	res.	e_{u_x}	e_M	res.	e_{u_x}	e_M	res.
Superposition, data loss only	6.8	3.7	0.067	14.6	11.0	0.229	8.1	9.9	0.094
Superposition + soft equilibrium penalty	6.5	3.7	0.063	14.0	11.0	0.219	8.3	9.7	0.088
Single combined load scale	12.2	5.4	0.078	24.5	15.0	0.240	15.1	10.2	0.097
Single combined load scale + Eucl. proj.	12.2	7.2	0	24.5	19.9	0	15.1	13.3	0

6.5. Uncertainty and FE referral

The ensemble spread ranks the frames by their actual error well on in-distribution frames and on unseen geometries (Spearman $\rho_s = 0.81$ and 0.92; Fig. 7 and Table 5), and the $\pm 2\sigma$ intervals of the end moments contain 92% of the PyNite values in both sets, close to the nominal 95%. It is much weaker for unseen load patterns ($\rho_s = 0.66$, coverage 67%): the networks agree with each other on these frames while being wrong together, which is the known failure mode of ensembles under distribution shift [19]. The label-free equilibrium residual is a weaker but non-trivial indicator ($\rho_s = 0.43$–0.59).

Figure 8 shows the 95th percentile of the moment error among the accepted predictions as a function of the fraction of frames referred to PyNite. On the ID set, referring the 20% most uncertain frames lowers it from 6.5% to 4.8% and catches 78% of the few frames (1.5%) with errors above 10%. On unseen geometries, where 35% of the frames exceed 10%, the same referral lowers the tail from 22.6% to 15.0%, and referring 50% brings it to about 9%, close to the oracle that knows the true errors. On unseen load patterns the

Table 5: Quality of the uncertainty estimate and FE referral. ρ_s: Spearman rank correlation between a frame-level score and the true moment error e_M; coverage: fraction of end moments within the ensemble mean $\pm 2\sigma$; ">10%": fraction of frames with $e_M > 10\%$; recall: share of those frames caught when the 20% highest-scoring frames are referred to PyNite; p95 after referral: 95th percentile of e_M over the accepted 80%; speed-up: average cost of PyNite divided by the cost of the workflow with 20% referral (Section 6.7).

Test set	ρ_s (spread)	ρ_s (residual)	Coverage (%)	>10% (%)	Recall (%)	p95 after (%)	Speed-up
In-distribution	0.81	0.57	92.3	1.5	78	4.8	3.7
Unseen geometries	0.92	0.59	92.2	34.8	56	15.0	3.8
Unseen loading	0.66	0.43	66.9	44.8	37	20.5	3.3

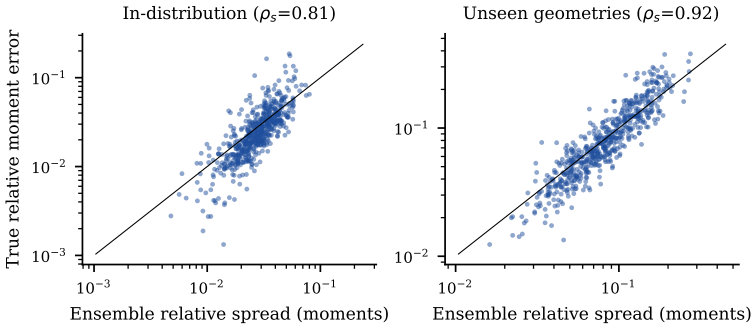


Figure 7: Frame-level ensemble spread of the end moments (Eq. 5) against the true relative moment error.

spread-based referral is little better than random (21.4%→20.5% at 20%), so the workflow cannot by itself detect this kind of shift; a domain check on the load pattern, which is known before any analysis, is the obvious complement.

6.6. Uncertainty-driven adaptive FE data generation

The adaptive study starts from 200 randomly chosen PyNite analyses and adds 200 per round up to 1,200, training a three-network ensemble (100 epochs) in every round; the whole study is repeated with two random seeds, and Fig. 9 shows the seed mean with the range of the two runs. Table 6 lists the errors at the final budget. Because these ensembles are smaller and trained on fewer frames for fewer epochs, their errors are higher than those of the main model in Table 2.

Two findings stand out. First, selecting the frames with the largest ensemble variance is *worse* than random sampling on all three test sets and for both seeds (median moment error 6.0% vs. 5.4% ID, 18.8% vs. 14.9% OOD-G

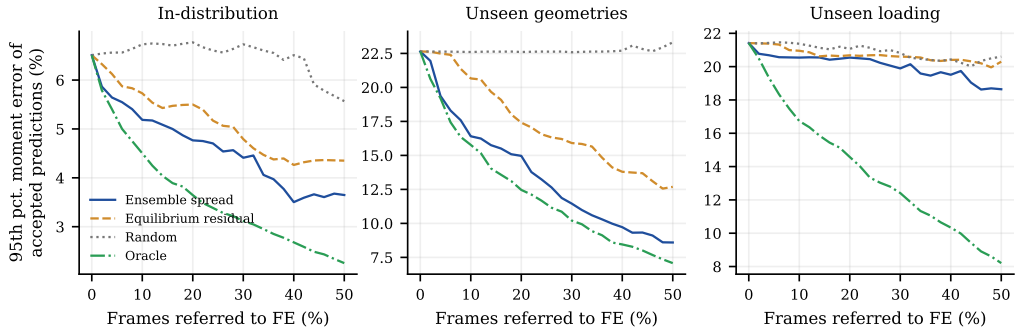


Figure 8: Risk–coverage curves: 95th percentile of the moment error over the accepted predictions when the highest-scoring frames are referred to PyNite, for the ensemble spread, the equilibrium residual, random referral and an oracle ranking by the true error.

Table 6: Errors (%) after 1,200 PyNite analyses for each acquisition rule (mean of two seeds): median and 95th percentile of the moment error and median horizontal displacement error.

	In-distribution			Unseen geometries			Unseen loading		
Acquisition	e_M	e_M (p95)	e_{u_x}	e_M	e_M (p95)	e_{u_x}	e_M	e_M (p95)	e_{u_x}
Random	5.4	13.3	9.3	14.9	36.2	16.4	11.8	25.4	10.5
Ensemble variance	6.0	14.9	12.2	18.8	49.8	24.0	12.5	28.7	12.6
Equilibrium residual	5.2	12.9	11.0	11.8	27.1	16.5	10.6	21.7	12.6

at 1,200 analyses). Greedy top-B selection by variance concentrates the budget on the most extreme corners of the design space, typically frames with very flexible members or unusual stiffness ratios, at the expense of coverage of the bulk of the distribution; a batch of 200 such frames is also highly redundant. Second, selecting the frames whose raw prediction is furthest from equilibrium is the best of the three rules for the member forces on every test set: at 1,200 analyses the median moment error is 5.2% (random 5.4%) on ID frames, 11.8% (14.9%) on unseen geometries and 10.6% (11.8%) on unseen loading, and the 95th percentile on unseen geometries falls from 36.2% to 27.1%. On unseen geometries the equilibrium-residual rule reaches the final error of random sampling after 600 analyses, i.e. with half the FE budget, and on unseen loading after 800. The gain is specific to forces: the displacement errors are similar to or slightly larger than with random sampling (Table 6), which is expected, since the residual measures the admissibility of the forces only. With two seeds, the differences between the equilibrium-

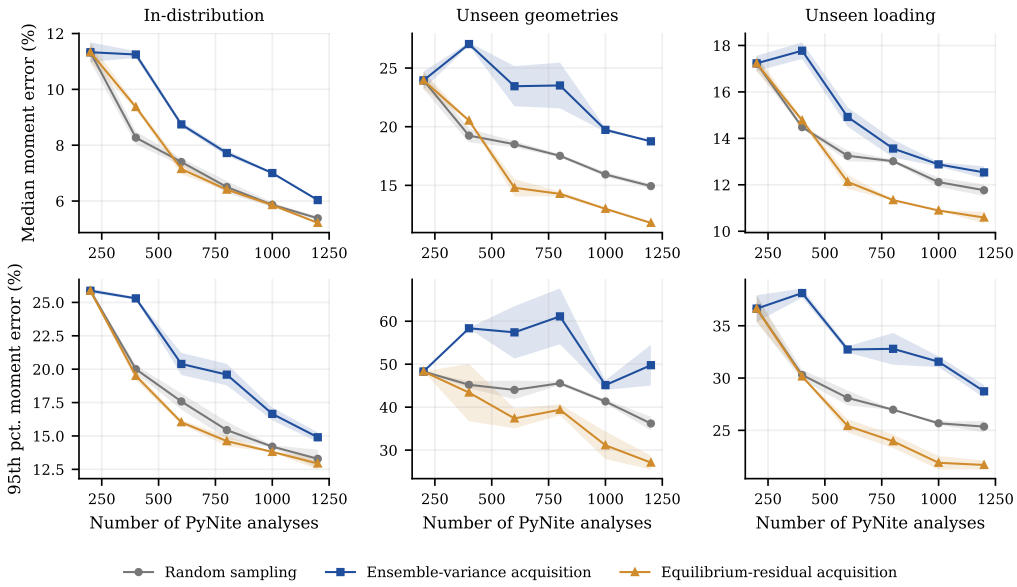


Figure 9: Moment error (median, top; 95th percentile, bottom) versus the number of PyNite analyses used for training, for three acquisition rules. Lines: mean of two seeds; bands: range of the two seeds. Predictions include the variance-weighted projection.

residual rule and random sampling on the ID set are within the seed-to-seed spread, whereas those on the unseen sets are consistent across both seeds.

6.7. Computational cost

Timings were measured serially on an otherwise idle 4-core CPU (Table 7). A PyNite analysis of one frame (two load cases, including model construction and result extraction) costs 37.5 ms for ID frames and 104.7 ms for the larger OOD-G frames. Evaluated in batches, the proposed surrogate costs 2.7 ms (ID), 7.0 ms (OOD-G) and 3.7 ms (OOD-L) per frame, including the Python feature construction (the largest part), the five-network ensemble and the projection, i.e. 10–15 times less than PyNite (Table 7); the projection itself takes 0.1–0.6 ms. For a single frame evaluated in isolation, the framework overhead of the networks dominates and the surrogate (17–82 ms) is not faster than PyNite. With 20% of the frames referred back to PyNite, the expected speed-up of the batched workflow drops to 3.3–3.8 (Table 5).

These savings must be set against the one-off cost of building the surrogate: training one network on the 3,000 training frames took 487 s, so

Table 7: Wall-clock time per frame (ms) on a 4-core CPU: PyNite analysis (mean of 200 frames), surrogate stages in batched evaluation, batched total, end-to-end time for one frame evaluated alone, and the batched speed-up without referral.

Test set	PyNite	Features	Ensemble	Projection	Batched total	Single frame	Speed-up
In-distribution	37.5	1.52	1.01	0.13	2.66	16.8	14.1
Unseen geometries	104.7	3.83	2.51	0.61	6.96	82.3	15.0
Unseen loading	35.1	2.48	1.05	0.15	3.67	27.9	9.6

the five-network ensemble costs about 41 min of CPU time, while the 3,000 PyNite analyses of the training set cost only about 2 min. With 20% referral, the surrogate therefore pays for itself only after roughly 9×10^4 ID frames (about 3×10^4 of the larger OOD-G frames). For first-order linear analysis of frames of this size, PyNite is already fast, and the surrogate is worthwhile only for very large numbers of repeated evaluations, for batched evaluation inside optimisation or uncertainty-propagation loops, or when the training cost is shared across many studies. The balance shifts in favour of the surrogate as the cost of a single FE analysis grows, but this paper does not test surrogates for the more expensive analyses (nonlinear, dynamic, or much larger models) where that would be the case.

7. Discussion

Answer to the research question. Within the design space it was trained on, the surrogate predicts member forces accurately enough to replace most repeated linear analyses at the level of preliminary design and screening: the median error of the end-moment field is 2.4%, the largest end moment of a frame is within 0.8% at the median, and the forces are exactly in equilibrium with the loads. Displacements are less accurate (median 4.5% for the horizontal displacement field, 4.9% for the governing drift). Outside the training space the median errors roughly triple and the 95th percentile of the moment error exceeds 20%; the surrogate is then usable only together with a mechanism that sends doubtful cases back to the FE solver. That mechanism works for unseen geometries, where the ensemble spread ranks the errors well, but not for unseen load patterns, where the ensemble is confidently wrong. The answer is therefore a qualified yes: the surrogate can reduce repeated FE analysis inside a validated domain and, with referral, for moderately larger frames, but it is not a substitute for FE analysis on load patterns that the training data did not cover.

What the constraints buy. The constraints that hold exactly by construction contribute in two different ways. Exact load-case superposition and dimensional scaling are an *accuracy* device: they remove the need to learn the dependence on the load intensities and on E, and they make the lateral and gravity responses equally well conditioned, which roughly halves the displacement errors (Table 4). The equilibrium projection is primarily an *admissibility* device: it guarantees that the forces are in equilibrium, which a soft penalty does not achieve, and with a variance-weighted metric it also improves the forces, strongly for the axial forces and modestly for the moments. With a poorly chosen metric the projection can degrade the moments, so the metric is an essential part of the method rather than a detail. The projection does not enforce compatibility or the constitutive law: the predicted displacements and forces are not guaranteed to be consistent with each other, and imposing both equilibrium and compatibility exactly would amount to solving the FE problem.

Uncertainty as a decision tool. The two uses of uncertainty behaved differently. At deployment, the ensemble spread is a good indicator of which frames to refer to FE analysis for in-distribution and geometrically extrapolated frames, but its calibration collapses under a change of load pattern (coverage of the $\pm 2\sigma$ intervals 67% instead of 92%). During training-set construction, greedy acquisition by ensemble variance was worse than random sampling, whereas acquisition by the equilibrium residual, a physics-based indicator that requires no labels, was the best rule for the member forces and halved the FE budget needed to reach a given accuracy on unseen geometries. A plausible explanation is that the residual is driven by the parts of the response that statics can check, whereas the variance also rewards the most atypical parameter combinations, which contribute little to the bulk of the test distribution. Batch-diversity and density-weighted acquisition rules, which are common remedies, were not tested.

Cost. The per-frame speed-up of the batched surrogate over PyNite is about 10–15, and 3.3–3.8 with 20% referral. Because a linear PyNite analysis of these frames takes only 30–100 ms, the one-off training cost of the ensemble (about 41 min of CPU time) is recovered only after tens of thousands of evaluations. This is an honest limitation of applying the method to cheap linear analyses and should be weighed against the specific use case; the accuracy, admissibility and uncertainty results do not depend on it.

Limitations. The study is restricted to planar, regular, multi-storey moment frames with prismatic members, rigid joints, uniformly distributed beam loads and floor-level lateral forces, analysed with first-order linear elasticity. Sections and loads were sampled independently, so many frames would not satisfy serviceability limits; the dimensionless formulation makes the relative errors invariant to this, but the parameter correlations of designed frames are not represented. All results come from one dataset, one training run of each main model and two seeds of the adaptive study, so small differences (for example between the two projection metrics on in-distribution frames) should not be over-interpreted. The adaptive study is retrospective: candidates were labelled in advance and their labels revealed only on request, which yields exact budget accounting but not the scheduling overheads of a live FE loop. Timings were measured on one CPU with Python implementations of both the surrogate and PyNite.

Outlook. Natural extensions are a mixed formulation that also enforces compatibility through the member flexibilities, extension to three-dimensional frames, braced systems and second-order (P-Δ) analysis, in which the FE cost is higher and superposition no longer holds exactly, explicit domain checks on the load pattern to complement the ensemble, and diversity-aware batch acquisition.

8. Conclusions

A mechanics-constrained surrogate modelling framework for the linear response of planar moment frames was developed and evaluated on 7,200 PyNite analyses. A graph network ensemble predicts nodal displacements and member basic forces in a dimensionless, load-case-separated form, so that superposition, homogeneity in the loads and in Young's modulus and the boundary conditions hold exactly; an equilibrium projection with an uncertainty-weighted metric makes the predicted forces statically admissible; and the uncertainty estimate decides which frames are analysed with PyNite during training and at deployment. The main conclusions are:

1. On in-distribution frames the surrogate reaches median errors of 2.4% for the end-moment field, 0.6% for the axial forces, 0.8% for the largest end moment and 4.5% for the horizontal displacements, four to six times lower than the best of five regression and ML baselines given identical features.

2. On unseen geometries and unseen load patterns the median moment errors are 7.5% and 8.7% (best baseline 22.1–22.5% and 16.8–17.3%), but the 95th percentiles exceed 20%.
3. The equilibrium projection reduces the out-of-balance force from 4–16% (raw ensemble) and up to 120% (baselines) to machine precision for every model; with a variance-weighted metric it also reduces the force errors, whereas a Euclidean metric can increase the moment errors. A soft equilibrium penalty is ineffective.
4. Exact load-case superposition roughly halves the displacement errors compared with a single load scale.
5. The ensemble spread ranks the frame errors well for in-distribution frames and unseen geometries (Spearman 0.81 and 0.92) and supports effective FE referral there, but it is poorly calibrated for unseen load patterns.
6. For adaptive data generation, acquisition by the equilibrium residual of the raw prediction gave the most accurate member forces and reached the accuracy of 1,200 random analyses on unseen geometries with 600 analyses, whereas acquisition by ensemble variance was worse than random sampling.
7. The batched surrogate is about 10–15 times faster than a PyNite analysis per frame, but the training cost is recovered only after tens of thousands of evaluations for analyses as cheap as those considered here.

The framework is therefore best viewed as a reliability layer that makes learned structural surrogates statically admissible and self-aware of their limits, rather than as a replacement for FE analysis.

Data and code availability

All scripts used to generate the PyNite dataset, train the models, run the experiments and produce every figure and table are in the `research/surrogate` folder of the PyNite fork (`https://github.com/karthice1/Pynite`). The dataset is regenerated deterministically from the seeds in `generate_dataset.py`.

CRediT authorship contribution statement

To be completed by the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to write the analysis code, run the experiments and draft the manuscript. All numerical results were produced by the released scripts. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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